



Yanzhou Mu

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Research direction: Software Testing

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EDUCATIONAL EXPERIENCE

- **Nanjing University** *Mentor: Zhenyu Chen, Chunrong Fang* **Profession:** Software Engineering
Software College, Nanjing City **Ph.D. Student** 2022.09-2026.06
- **Tianjin University** *Mentor: Zang Wang, Junjie Chen* **Profession:** Software Engineering
Department of Intelligence and Computing **Master** 2019.09-2022.01
- **Nantong University** *Mentor: Xiang Chen* **Profession:** Computer Science and Technology
School of computer science and technology **Bachelor** 2015.09-2019.06

PUBLICATIONS

2.1 First-Author Publications

- **Yanzhou Mu**, Juan Zhai, Chunrong Fang, Xiang Chen, Zhixiang Cao, Peiran Yang, Yinglong Zou, Tao Zheng, Zhenyu Chen. DevMuT: Testing Deep Learning Framework via Developer Expertise-Based Mutation. In Proceedings of IEEE/ACM Automated Software Engineering Conference (ASE 2024). ACM, Sacramento, California, United States. CCF-A
- **Yanzhou Mu**, Juan Zhai, Chunrong Fang, Xiang Chen, Zhixiang Cao, Peiran Yang, Kexin Zhao, An Guo and Zhenyu Chen. Improving Deep Learning Framework Testing with Model-Level Metamorphic Testing. International Symposium on Software Testing and Analysis (ISSTA 2025), Thondheim, Norway, 2025. CCF-A
- **Yanzhou Mu**, Rong Wang, Juan Zhai, Chunrong Fang, Xiang Chen, Peiran Yang, Zhixiang Cao, Ruixiang Qian, Shaoyu Yang, and Zhenyu Chen. Deep Learning Framework Testing via Model Mutation: How far are we? IEEE Transactions on Software Engineering, 2025, CCF-A
- **Yanzhou Mu**, Rong Wang, Juan Zhai, Chunrong Fang, Xiang Chen, Jiacong Wu, An Guo, Jiawei Shen, Bingzhuo Li, and Zhenyu Chen. Understanding LLM-Centric Challenges for Deep Learning Frameworks: An Empirical Analysis. ACM Transactions on Software Engineering and Methodology, 2026, CCF-A
- 沐燕舟, 王赞, 陈翔, 陈俊洁, 赵静珂, 王建敏. 采用多目标优化的深度学习测试优化方法 [J]. 软件学报, 2022, 33(07):2499-2524. DOI:10.13328/j.cnki.jos.006583.
- **Yanzhou Mu**, Zan Wang, Xiang Chen, Junjie Chen, Jingke Zhao, and Jianmin Wang. Deep learning test optimization method using multi-objective optimization. International Journal of Software & Informatics, 12(4), 2022. (上一篇文章于 2021 年度中国软件大会专刊论坛报告后, 被作为 5 篇推荐文章翻译后发表于此刊)
- **Yanzhou Mu**, Zan Wang, Shuang Liu, Jun Sun, Junjie Chen, and Xiang Chen. HARS: Heuristic-enhanced adaptive randomized scheduling for concurrency testing. In 2021 IEEE 21st International Conference on Software Quality, Reliability and Security (QRS), pages 219–230. IEEE, 2021.

2.2 Co-authored Publications

- Xiang Chen, **Yanzhou Mu**, Ke Liu, Zhanqi Cui, and Chao Ni. Revisiting heterogeneous defect prediction methods: How far are we? Information and Software Technology, 130:106441, 2021. **Co-first Author.**
- Chunyu Zhao, **Yanzhou Mu**, Xiang Chen, Jingke Zhao, Xiaolin Ju, and Gan Wang. Can test input selection methods for deep neural network guarantee test diversity? a large-scale empirical study. Information and Software Technology, 150:106982, 2022. **Co-first Author.**

- Xiang Chen, **Yanzhou Mu**, Yubin Qu, Chao Ni, Meng Liu, Tong He, and Shangqing Liu. Do different cross-project defect prediction methods identify the same defective modules? *Journal of Software: Evolution and Process*, 32(5):e2234, 2020. **Co-first Author**.

2.3 Earlier Publications

- Hao Li, Xiaohong Li, Xiang Chen, Xiaofei Xie, **Yanzhou Mu**, and Zhiyong Feng. Cross-project defect prediction via asttoken2vec and blstm-based neural network. In *2019 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2019.
- An Guo, Shuoxiao Zhang, Enyi Tang, Xinyu Gao, Haomin Pang, Haoxiang Tian, **Yanzhou Mu**, Wu Wen, Chunrong Fang, Zhenyu Chen. When Autonomous Vehicle Meets V2X Cooperative Perception: How Far Are We? *The 40th IEEE/ACM International Conference on Automated Software Engineering*, November 16-20, 2025, Seoul, South Korea. CCF-A
- Yuan Xiao, Yuchen Chen, Jiaming Wang, Wei Song, Jun Sun, Shiqing Ma, **Yanzhou Mu**, Juan Zhai, Chunrong Fang, Jin Song Dong, Zhenyu Chen. Train in Vain: Functionality-Preserving Poisoning to Prevent Unauthorized Use of Code Datasets. *Findings of the Association for Computational Linguistics*, 2026, CCF A)

2.4 Manuscripts Under Major Revision

- Yilong Zou, Juan Zhai, Chunrong Fang, **Yanzhou Mu**, Jiawei Liu and Zhenyu Chen. Deep Learning Framework Testing via Heuristic Guidance Based on Multiple Model Measurements, *ACM Transactions on Software Engineering and Methodology*, 2025, CCF-A, **Major revision**.
- An Guo, Xinyu Gao, Chunrong Fang, Haoxiang Tian, Weisong Sun, **Yanzhou Mu**, Shuncheng Tang, Lei Ma, Xiapu Luo and Zhenyu Chen. Generate Realistic Test Scenes for V2X Communication Systems. *ACM Transactions on Software Engineering and Methodology*, 2025, CCF-A, **Major revision**.

PROJECT EXPERIENCE

- **MindSpore Network Generalization Testing Technology | (Completed, Student Leader)**
 - Developed a network structure generalization tool based on the domestic deep learning framework MindSpore, which extracts the feature factors, ranges, and constraints of various mainstream CV and NLP network architectures. The tool modifies the network structure and performs testing to detect defects. **The project results were an essential part in supporting the collaboration team to win the Huawei 2023 Innovation Testing Application Award.**
- **Research on Key Mutation Testing Techniques for DL Frameworks | (Ongoing, Student Leader)**
 - Designed mutation rules that simulate actual user development operations to generate test models aligned with practical scenarios. Evaluated the diversity and sufficiency of test data by integrating the framework’s static and dynamic features. Employed heuristic methods to guide mutation testing, and combined experience replay to improve testing efficiency and quality. **Project results were published at the CCF-A conference ASE’24 and a patent application for an invention is currently underway.**
- **Fast Detection, Localization, and Patch Generation for Concurrency Defects | (Completed, Project Participant)**
 - Designed a heuristic scheduling testing method to optimize Java concurrent programs, enhancing detection efficiency and effectiveness. **Project results were published at the CCF-C conference QRS’21 and granted an invention patent.**
- **LLM Generalization Testing Tool for MindSpore | (Ongoing, Student Leader)**

- Developed a component-based generalization testing tool for large-scale models on MindSpore, enabling automated mutation and recombination of model structures (e.g., layers, shapes, and parameters) under defined constraints. Built cross-framework validation pipelines by generating equivalent training scripts in PyTorch and conducting comparative analysis to detect training failures and accuracy anomalies. Designed iterative strategies to identify fault-inducing factors and support multi-dimensional issue diagnosis across functionality, performance, and numerical accuracy.

PATENT

- **Yanzhou Mu**; Zan Wang; Shuang Liu. A heuristic rule-based concurrent adaptive random testing method. Patent No: CN113468047A.
- Junjie Chen; **Yanzhou Mu**; Zan Wang; Jianmin Wang; Jiao Jia. A multi-objective optimization-based method for selecting deep learning test inputs. Patent No: CN114721934A.
- Zhenyu Chen; Yidong Ke; Jiawei Liu; **Yanzhou Mu**. A neural network fuzz testing method guided by mutation image entropy values. Patent No: CN117218051A.
- Yongping Liu; Zhenyu Chen; Chunrong Fang; **Yanzhou Mu**; Ruifa Luo; Shuiyong Du. Mutation testing method, device, and computer equipment based on neuron characteristics of intelligent traffic models. Patent No: CN118689772A.
- Yongping Liu; Zhenyu Chen; Chunrong Fang; **Yanzhou Mu**; Ruifa Luo; Shuiyong Du. Domain-specific mutation testing method, device, and computer equipment based on intelligent traffic V2X models. Patent No: CN118689771A.
- Zhijin Guan; Haiying Ma; Hehe Gu; Zongyuan Zhang; **Yanzhou Mu**; Jing Zhao; Meng Liu. A quantum circuit image recognition method. Patent No: CN106997462A.